

Math 2551 Worksheet: Lines and Planes

1. Find an equation for
 - (a) the line through point $P = (1, 2, -1)$ and point $Q = (-1, 0, 1)$.
 - (b) the line through $(0, -7, 0)$ perpendicular to the plane $x + 2y + 2z = 13$.
 - (c) the line in which the planes $3x - 6y - 2z = 3$ and $2x + y - 2z = 2$ intersect.
2. Find a vector in the direction of the line of intersection ℓ of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Find a plane which goes through $(2, 1, -1)$ and is perpendicular to ℓ (and thus both planes).
3. Find 2 planes that are not parallel that both contain the points $P(1, -1, 1)$, $Q(3, 2, 0)$, and $R(5, 5, -1)$. When will 3 distinct points NOT determine a unique plane?
4. Find the point where the line $\mathbf{r}(t) = \langle 2, 3 + 2t, 1 + t \rangle$ intersects the plane $2x - y + 3z = 6$.
5. How can you tell when two planes $A_1x + B_1y + C_1z = D_1$ and $A_2x + B_2y + C_2z = D_2$ are parallel? Perpendicular? Justify your answer.
6. Find the point at which the lines $\ell_1(t) = \langle 2, 3, 1 \rangle + \langle 1, -1, 1 \rangle t$ and $\ell_2(t) = \langle 2, 1, -2 \rangle t + \langle 6, 2, 1 \rangle$ intersect.